

Claude Certified Architect Foundation - Labs

Six independent, offline-first labs aligned to the CCAR-F exam domains. Each folder contains its own guide, fixtures, implementation and verifier.

- [Lab 1: Customer Support Resolution Agent](#) - Build a bounded agent loop that enforces a refund prerequisite, normalizes tool output, checkpoints state, and escalates high-impact ambiguity.
- [Lab 2: Claude Code Team Workspace](#) - Create an auditable instruction hierarchy with path-specific rules, a reusable skill, an MCP server declaration, and a plan-versus-execute decision gate.
- [Lab 3: Multi-Agent Research System](#) - Delegate bounded research tasks to isolated workers, propagate structured errors, and synthesize claims without losing provenance or uncertainty.
- [Lab 4: Developer Productivity Agent](#) - Explore a codebase with a narrow Glob-Grep-Read sequence, record findings in an external scratchpad, and avoid whole-repository ingestion.
- [Lab 5: Claude Code CI Review](#) - Create a deterministic CI review contract with JSON Schema, per-file review, integration review, false-positive criteria, and a severity-based exit gate.
- [Lab 6: Structured Data Extraction](#) - Extract invoice data in batches, validate syntax and semantics, retry with field-level feedback, and route low-confidence records to human review.